

OS-MindAgent: Doubts & Revision Summary

Subject: Operating Systems (Course T7510) | Generated: September 15, 2026 - 12:24 PM | Filter: All

Total Doubts Resolved: 2

#1 | [Unit 2] (Confidence: 94%) — 2026-09-15 12:24:29

Q: What is the difference between Round Robin and FCFS scheduling?

A: ### ■ Core Concept: What is the difference between Round Robin and FCFS scheduling?

****Unit Classification:**** Unit 2

- **Definition**:** In Operating Systems, this concept handles how system resources (CPU, Memory, Disk) are assigned and managed efficiently.
- **Key Steps**:**
 - The OS tracks resource states in kernel data structures (e.g., Process Control Block or Page Tables).
 - Decisions are made based on preemptive or non-preemptive policies.
 - Interrupts or system calls signal transitions between User mode and Kernel mode.
- **Example**:**
 - In C/Linux programming: `fork()` creates a child process, `exec()` loads a program, and `wait()` synchronizes parent and child.
- **Exam Tip**:** Always mention state transitions, PCB attributes, and execution context when writing answers for Unit 2.

(Note: Add your `GROQ_API_KEY` in `.env` or `config.py` for live AI explanations via Groq Llama-3.3-70B).

#2 | [Unit 2] (Confidence: 94%) — 2026-09-15 12:23:27

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